

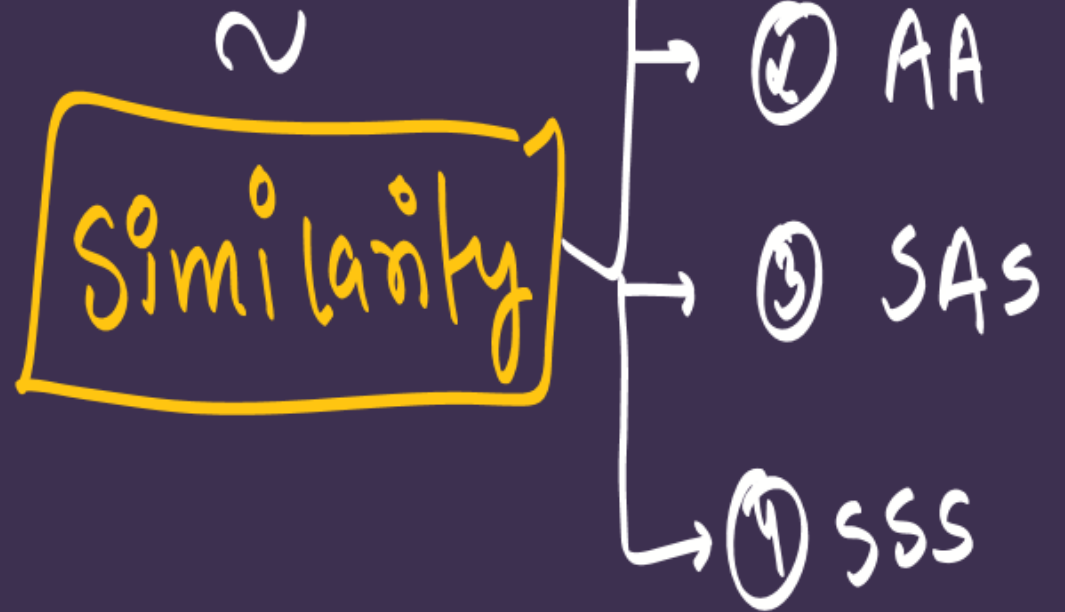
[Aim: 100/100 in Maths]

आरंभ

TRIANGLES

Lecture #4

ATK²:



given

$$\triangle \underline{ACB} \sim \triangle \underline{QPR}$$

$$\angle A = \angle Q$$

$$\angle C = \angle P$$

$$\angle B = \angle R$$

$$\frac{AC}{QP} = \frac{CB}{PR} = \frac{AB}{QR}$$

CPST

#LP: If in $\triangle ABC$ and $\triangle DEF$, if $\underline{AB} = \underline{BC}$ then they will be similar, when
DE FD

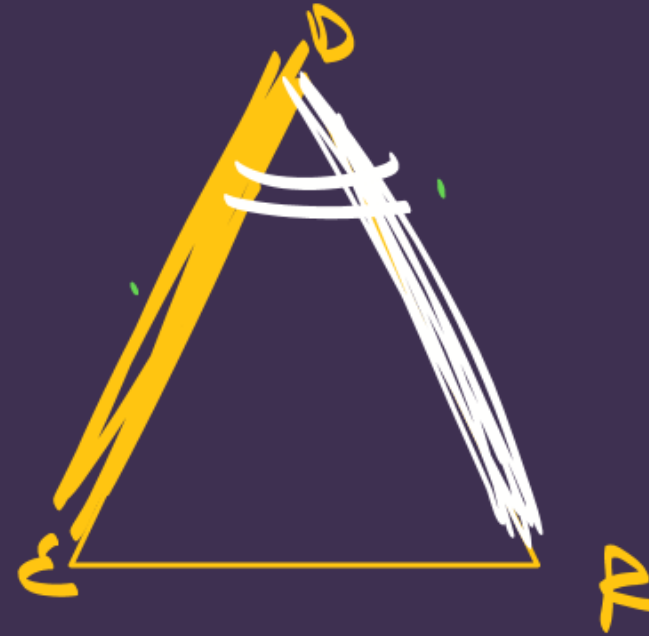
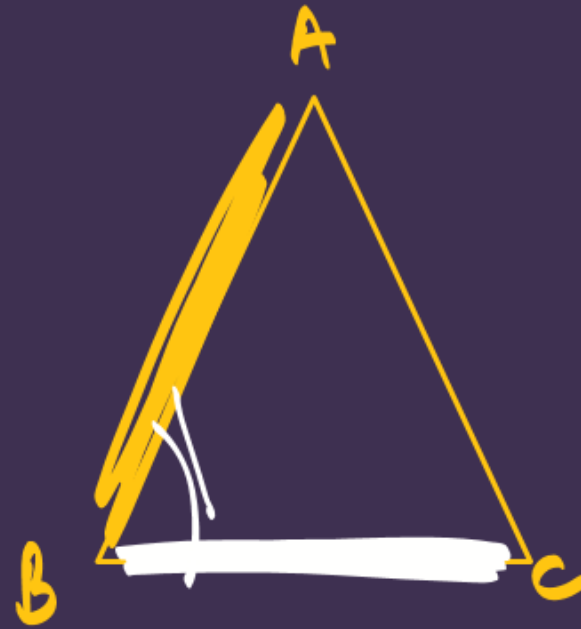
A. $\angle B = \angle E$

B. $\angle A = \angle D$

C. $\angle B = \angle D$

D. $\angle A = \angle F$

33.61%
32.02%
29.63%
4.1%



SAS

included angle

#LP: If $\triangle ABC \sim \triangle EDF$, then which of the following is not true?

A) $BC \cdot EF = AC \cdot FD$

B) $AB \cdot EF = AC \cdot DE$

☒ C) $BC \cdot DE = AB \cdot EF$ 38.5%

D) $BC \cdot DE = AB \cdot FD$

~~(A)~~ $\frac{BC}{FD} = \frac{AC}{EF}$

~~(B)~~ $\frac{AB}{DE} = \frac{AC}{EF}$

(C) $\frac{BC}{EF} = \frac{AB}{DE}$ X

#LP: If is given that $\triangle ABC \sim \triangle DFE$, $\angle A = 30^\circ$, $\angle C = 50^\circ$, $AB = 5$ cm, $AC = 8$ cm and $DF = 7.5$ cm. Then, which of the following is true?

A) $DE = 12$ cm, $\angle F = 50^\circ$ ✗

B) $DE = 12$ cm, $\angle F = 100^\circ$ ✓

C) $EF = 12$ cm, $\angle D = 100^\circ$ ✗

D) $EF = 12$ cm, $\angle D = 30^\circ$ ✓

$\angle A = \angle D = 30^\circ$

$\angle B = \angle F = 100^\circ$

$\angle C = \angle E = 50^\circ$

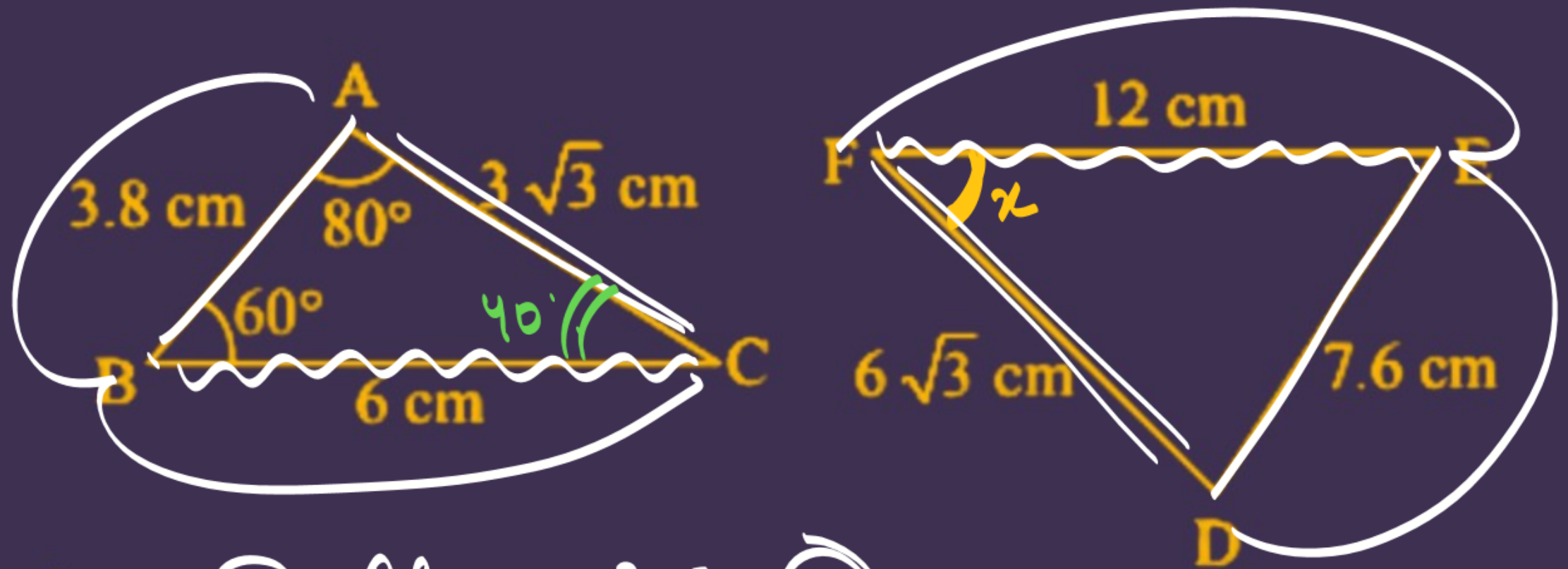
$$\frac{AB}{DF} = \frac{BC}{FE} = \frac{AC}{DE}$$

$$\frac{5}{7.5} = \frac{BC}{FE} = \frac{8}{DE}$$

$$\frac{5}{7.5} = \frac{8}{DE}$$

$DE = 12$

#LP: Find $\angle F$



SSS $\Rightarrow \Delta ABC \sim \Delta DEF$

$\angle F = \angle C = 40^\circ$

$$\frac{AB}{DE} = \frac{3.8}{7.6} = \frac{1}{2}$$
$$\frac{BC}{FE} = \frac{6}{12} = \frac{1}{2}$$
$$\frac{AC}{FD} = \frac{3\sqrt{3}}{6\sqrt{3}} = \frac{1}{2}$$

#(K3B) Q $BC \parallel DE$

(i)

$$\frac{AE}{EC} =$$

by BPT

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{2}{3} = \frac{AE}{EC}$$

(ii)

$$\frac{DE}{BC} = ?$$

No use of BPT

similarity ✓

Sol:- In $\triangle ADE$ & $\triangle ABC$

$$\begin{aligned} \angle 1 &= \angle 2 \quad \left\{ \begin{array}{l} \text{opp.} \\ \text{angles} \end{array} \right\} \\ \angle 3 &= \angle 4 \quad \left\{ \begin{array}{l} \text{alt.} \\ \text{angles} \end{array} \right\} \end{aligned}$$

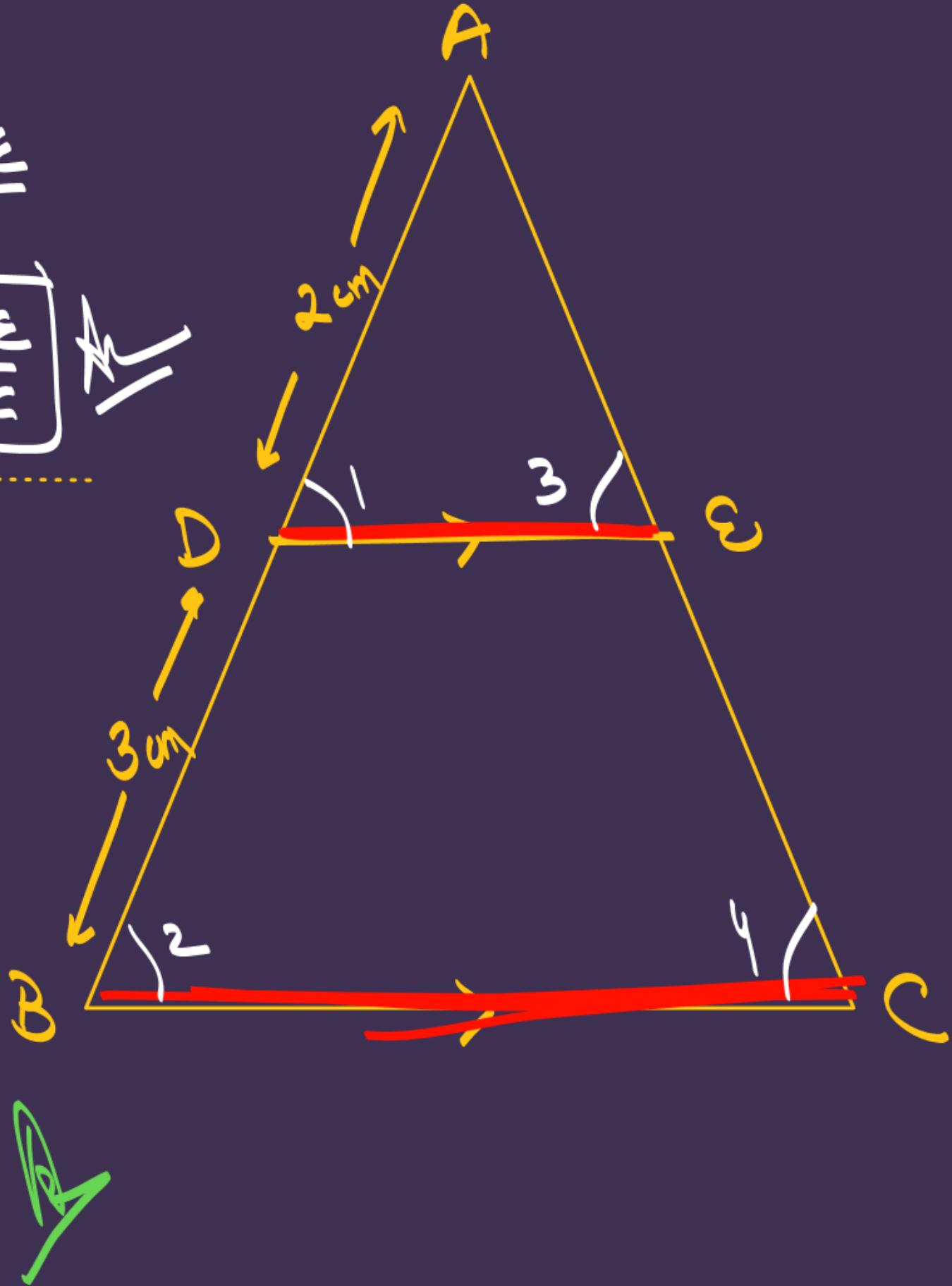
AA $\Rightarrow$

$$\triangle ADE \sim \triangle ABC$$

cpst

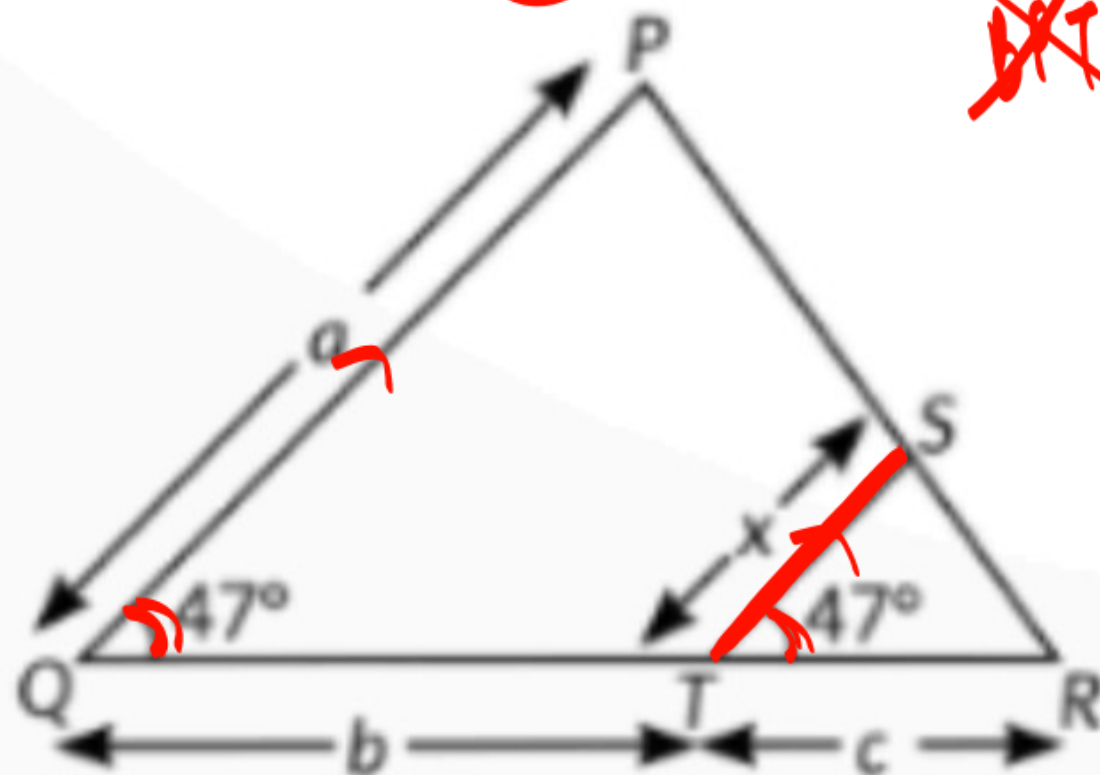
$$\frac{AD}{AB} = \frac{DE}{BC}$$

$$\frac{2}{5} = \frac{DE}{BC}$$



#LP:

In the given figure, x expressed in terms of a, b, c , is



~~similarity~~
similarity.
 $\triangle RST \sim \triangle RPQ$

$$\frac{RT}{RQ} = \frac{ST}{PQ}$$

$$\frac{c}{c+b} = \frac{x}{a}$$

$$\boxed{\frac{ac}{b+c} = x}$$

(a) $x = \frac{ab}{a+b}$

48.8% ✓

(b) $x = \frac{ac}{b+c}$

(c) $x = \frac{bc}{b+c}$

(d) $x = \frac{ac}{a+c}$

#LP: In $\triangle ABC$, DE is parallel to base BC, with D on AB and E on AC. If $AD = 2$, find BC . [CBSE 2002]

DB 3

DE

reciprocal

$$\frac{AD}{AB} = \frac{2}{2+3} = \frac{2}{5}$$

$$\frac{AB}{AD} = \frac{5}{2}$$

similarity

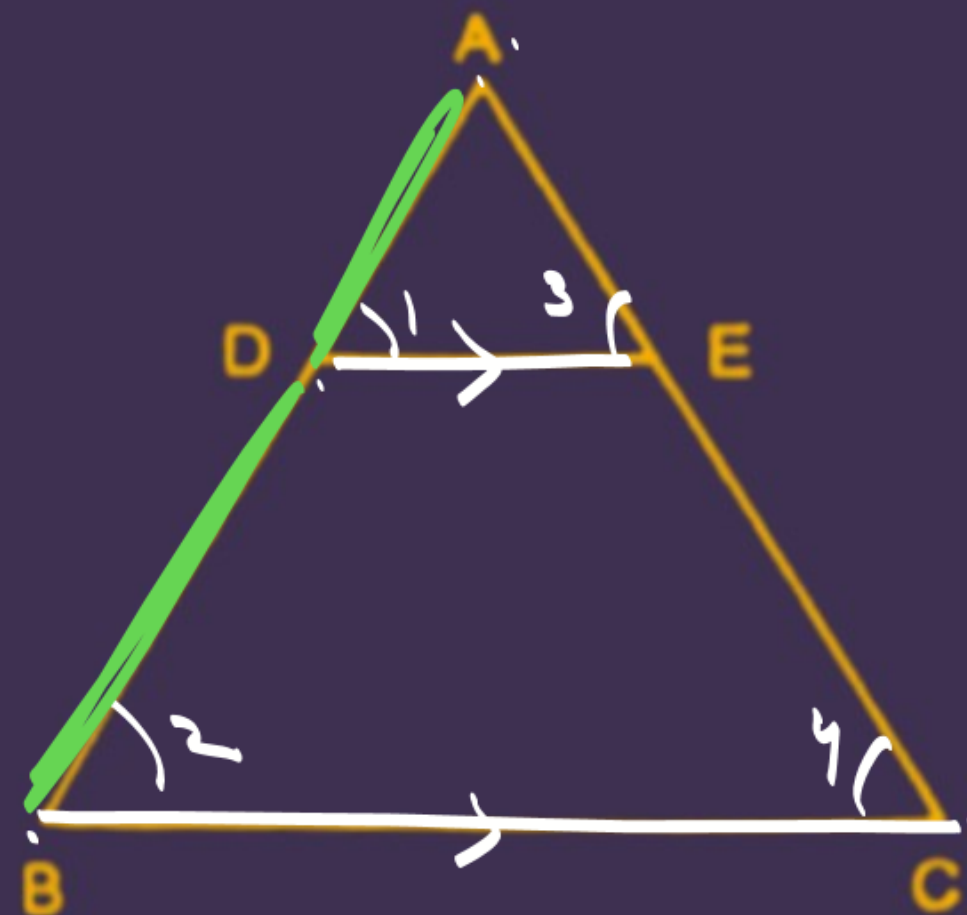
$$\begin{cases} \angle 1 = \angle 2 \\ \angle 3 = \angle 4 \end{cases} \text{AA} \Rightarrow$$

$$\triangle ADE \sim \triangle ABC$$

reciprocal

$$\frac{AD}{DB} = \frac{DE}{BC}$$

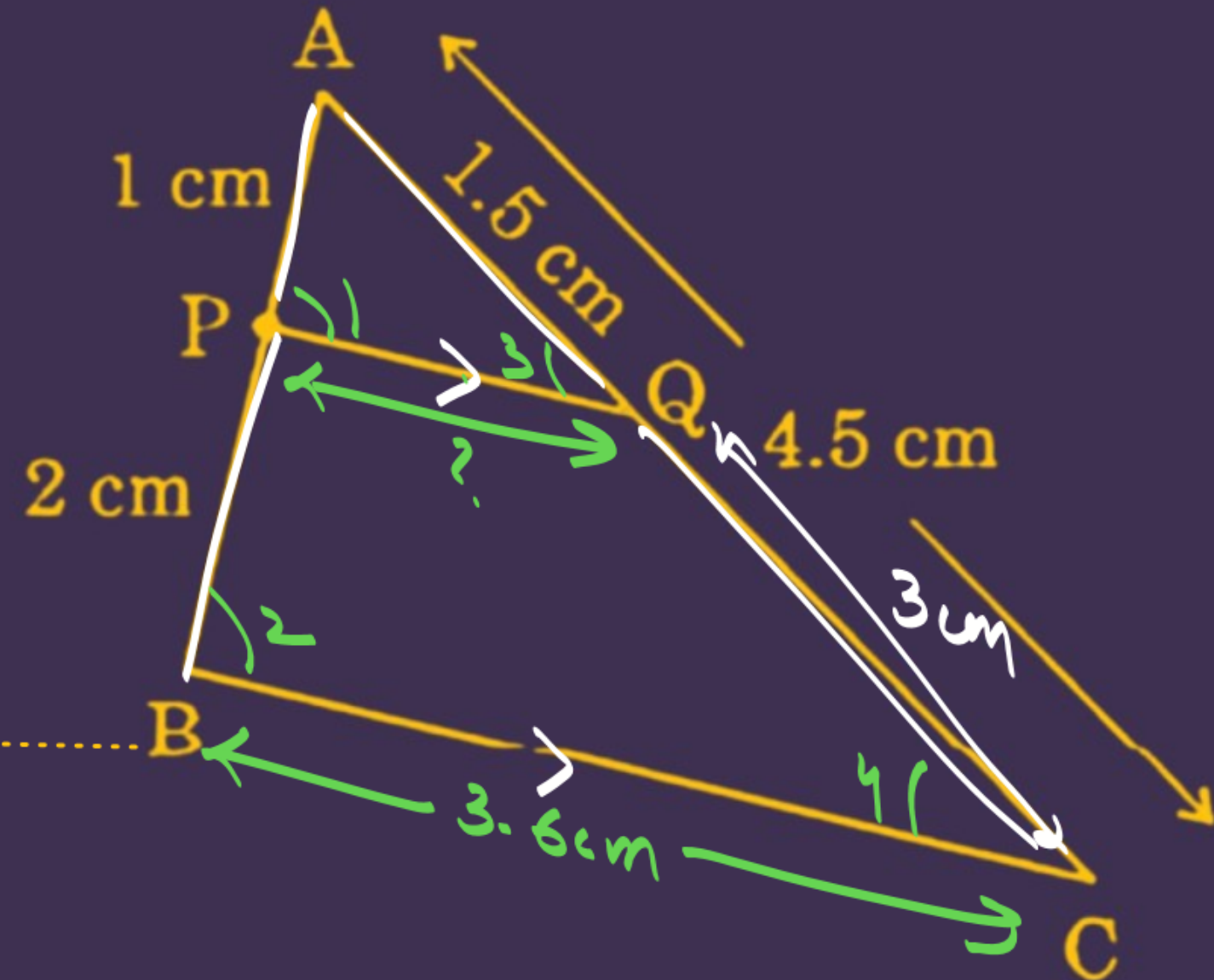
$$\frac{2}{3} = \frac{DE}{BC}$$



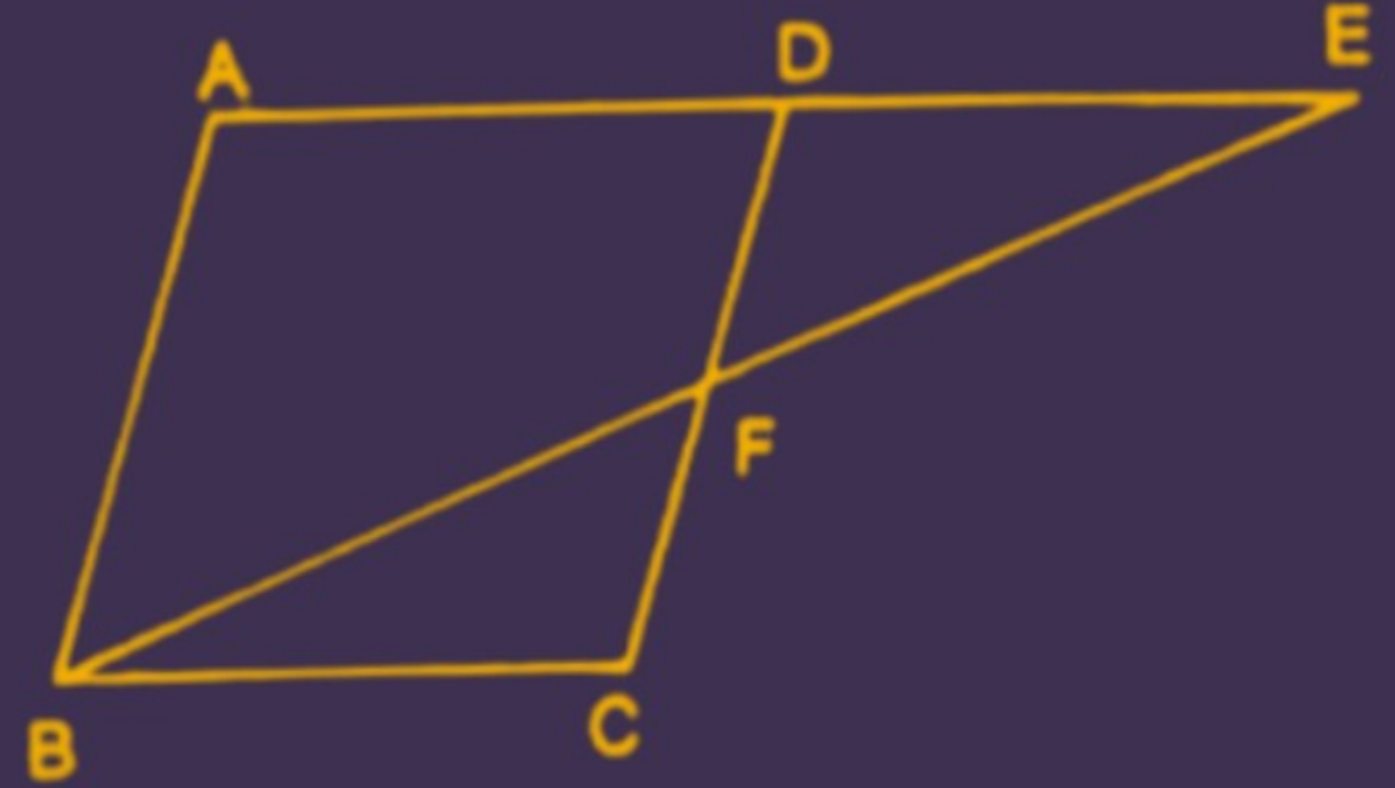
#LP: In the adjoining figure, $AP = 1 \text{ cm}$, $BP = 2 \text{ cm}$, $AQ = 1.5 \text{ cm}$ and $AC = 4.5 \text{ cm}$. Prove that $\triangle APQ \sim \triangle ABC$. Hence, find the length of PQ , if $BC = 3.6 \text{ cm}$. [CBSE 2025]

$\frac{AP}{PB} = \frac{1}{2} \quad \bigg| \quad \frac{AQ}{QC} = \frac{1.5}{3} \Rightarrow \frac{1}{2}$
 $\frac{AP}{PB} = \frac{AQ}{QC}$
 converse of BPT $\Rightarrow PQ \parallel BC$

$\text{corr. } \angle s \left[\begin{matrix} \angle 1 = \angle 2 \\ \angle 3 = \angle 4 \end{matrix} \right] \text{ (AA)} \Rightarrow \triangle APQ \sim \triangle ABC$



#LP: E is a point on side AD produced of a parallelogram ABCD and BE intersects CD at F. Prove that $\triangle ABE \sim \triangle CFB$.

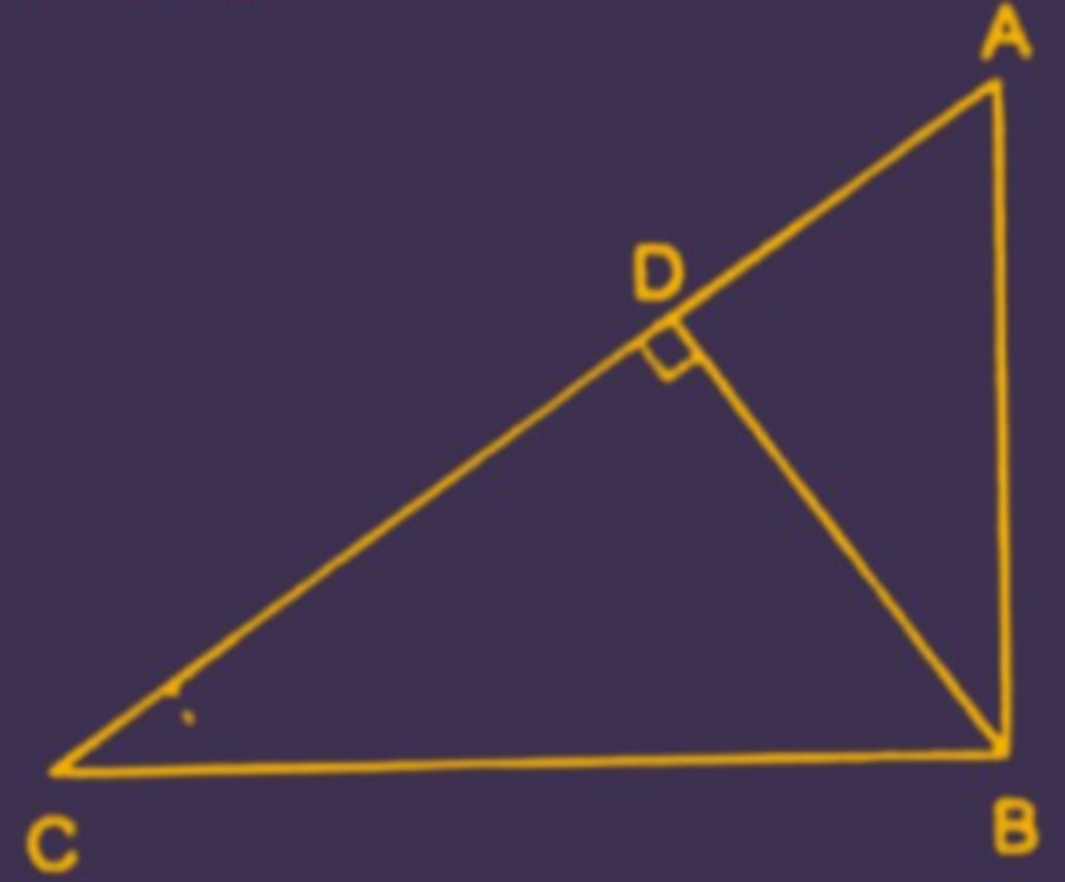


#LP: A right triangle ABC right angled at B. $BD \perp AC$. [CBSE 2009]

To prove :

i. $AB^2 = AD \times AC$

ii. $BC^2 = CD \times AC$

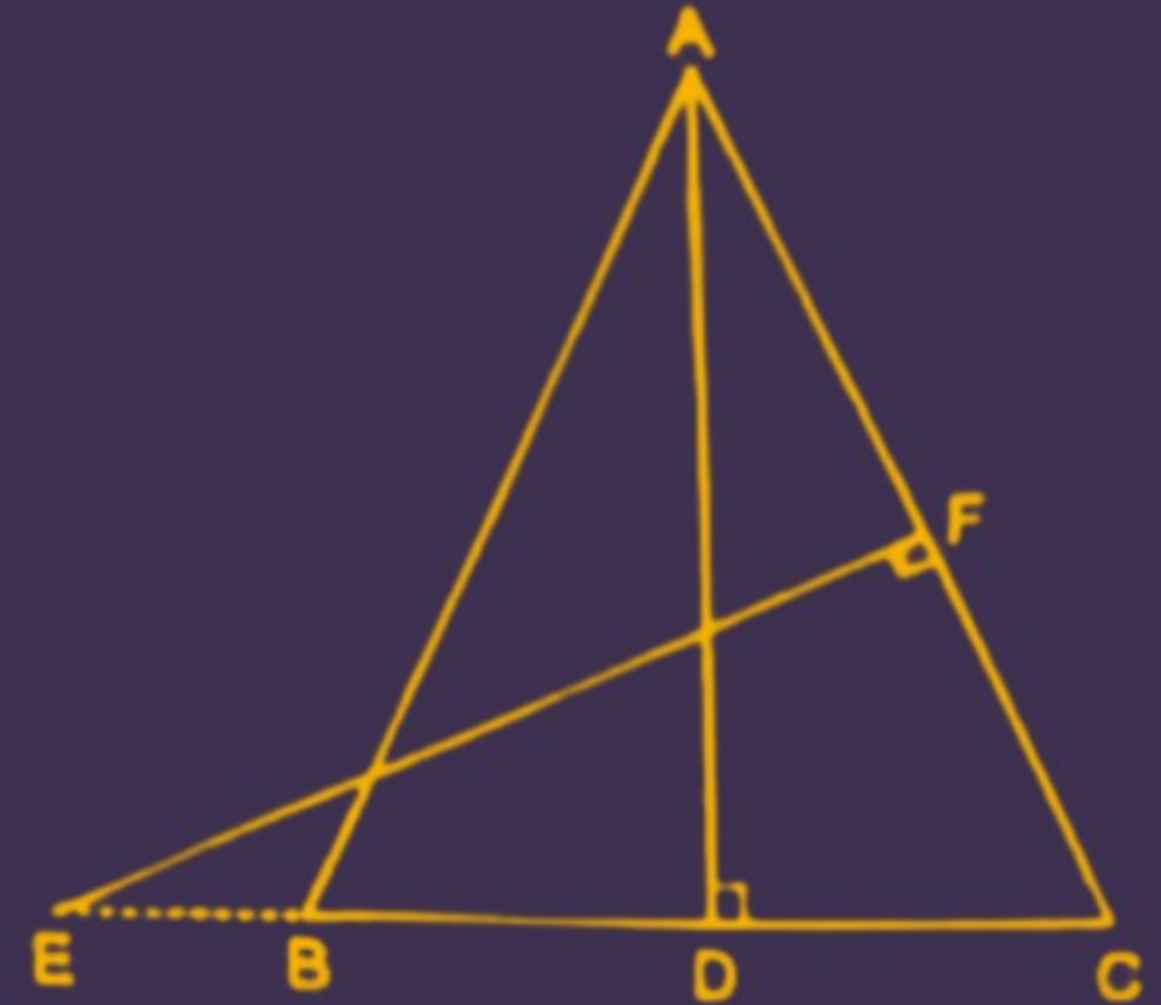


#LP: In Fig, E is a point on side CB produced of an isosceles triangle ABC with $AB = AC$. If $AD \perp BC$ and $EF \perp AC$, prove that

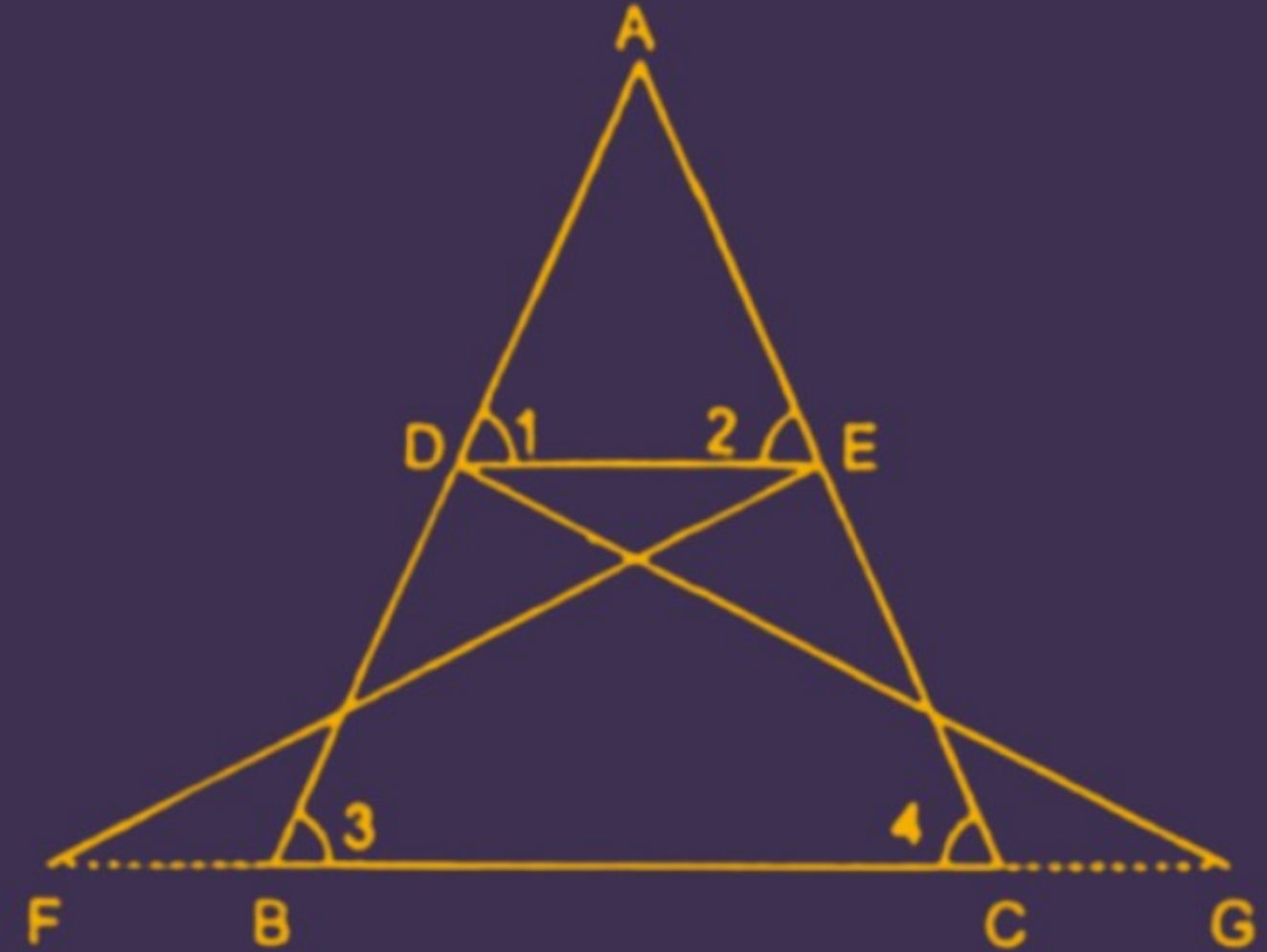
(i) $\triangle ABD \sim \triangle ECF$

(ii) $AB \times EF = AD \times EC$.

[CBSE 2010]



#LP: In the given figure , $\triangle FEC \cong \triangle GBD$ and $\angle 1 = \angle 2$, prove that $\triangle ADE \sim \triangle ABC$.



Thank you Goodies !!

